

As previously discussed with my UMGC AI bootcamp mentor, interactions with the ChatGPT API are more than sufficient for implementing my math chatbot capstone project. Presently my project prototype (built and executed in a Jupyter Notebook) uses TF-IDF and cosine similarity to implement rudimentary cross-attention between student answers to math questions from my math chatbot, teacher inputs to the chatbot in a JSON file, and responses from the ChatGPT API.

But it is possible that cross-attention in my project can be improved by exploring classifiers or regressors in scikit-learn, or deep neural networks in keras, to better determine if inputs of student reasoning are really relevant to a math problem and how to best choose different ChatGPT responses to a student's inputs. The changes made would be isolated/localized to the `get_content_similarity()` function in my present math chatbot prototype.

Training data for such a model could be the fields "question" and "conversation" (for X) and "self-correctness" (for y) in the math conversation dataset at:
<https://github.com/eth-nlped/mathdial>